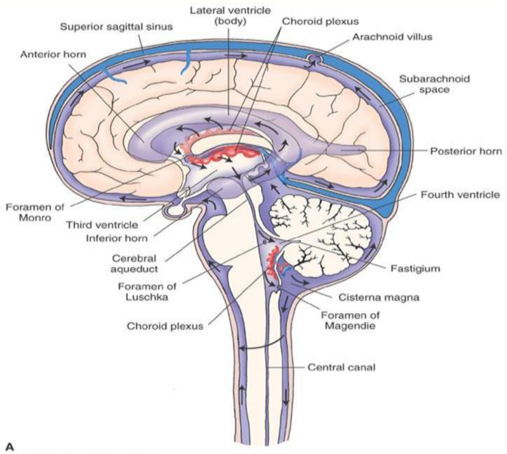
2026

Basic Principles of Medicine II
Module: Nervous System & Behavioral Sciences
Lecture No: NB 06

Transport within the CNS

Dr Esther Johnson



Note

- **Blue slides are the written explanation of the previous slide. Please do not ignore these slides while preparing for exams. The blue slides may not be shown during the lecture, however, it’s an integral part of the lecture material.**

Recommended Reading

Siegel and Sapru, Essential Neuroscience, 4th edition, © Wolters Kluwer Health, Inc. (2019), Chapter 3

Online link:

https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=196986286&bookid=2482

Lecture objectives

SOM.MK.I.BPM2.1.NB.1.NSCI.0061 List the fluid compartments in the CNS

SOM.MK.I.BPM2.1.NB.1.NSCI.0529 List the functions and constituents of the CSF

SOM.MK.I.BPM2.1.NB.1.NSCI.0530 Identify the typical sites for lumbar puncture in the adult

SOM.MK.I.BPM2.1.NB.1.NSCI.0062 Discuss principles of measurement and pathological elevation of intracranial pressure

SOM.MK.I.BPM2.1.NB.1.NSCI.0531 Describe the blood brain barrier (BBB) and its transport mechanisms

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SOM.MK.I.BPM2.1.NB.1.NSCI.0064 Describe the role of the choroid plexus in CSF formation and of the arachnoid villi in CSF drainage

SOM.MK.I.BPM2.1.NB.1.NSCI.0065 Describe transfer of solute and water across barriers in the central nervous system

SOM.MK.I.BPM2.1.NB.1.NSCI.0066 Correlate composition of the CSF with disease processes

SOM.MK.I.BPM2.1.NB.1.NSCI.0067 Identify the causes of non-communicating and communicating hydrocephalus

SOM.MK.I.BPM2.1.NB.1.NSCI.0068 Describe pathways for movement of fluid and solutes among different compartments in the CNS

SOM.MK.I.BPM2.1.NB.1.NSCI.0069 Identify the causes of brain edema, other than hydrocephalus

CNS Fluid Distribution

Total vol ~ 1200 ml

IC fluid ~ 720 ml

EC fluid ~ 480 ml

Blood Plasma 70 ml	Interstitial Fluid 260 ml	CSF 90-140 ml

CSF

- Produced by secretion via choroid plexus
- Circulates within a defined anatomical space
  - Ventricles, central canal (bulbar and spinal), subarachnoid space

CSF Functions

- Reduces traction on nerves and blood vessels because of buoyancy effect.

  - Buoyancy: weight of the brain reduced from ~1400 gm to ~50 gm, decreasing pressure on basal structures

- Cushioning and protection of the brain (reducing chances for contact with the skull)
- Removal of metabolites from the brain
- Provides a stable ionic environment

Composition of Normal Serum and Cerebrospinal Fluid

Constituent	Serum	CSF
Protein (g/L)	60–78	0.15–0.45
Glucose (mmol/L)	3.9–5.8	2.2–3.9
Ca^{2+} (mmol/L)	2.1–2.5	1–1.35
K^+ (mmol/L)	4–5	2.8–3.2
Na^+ (mmol/L)	136–146	147–151
Cl^- (mmol/L)	98–106	118–132
Mg^{2+} (mmol/L)	0.65–1.05	0.78–1.26
pH	7.41	7.33
HCO3-	24	22

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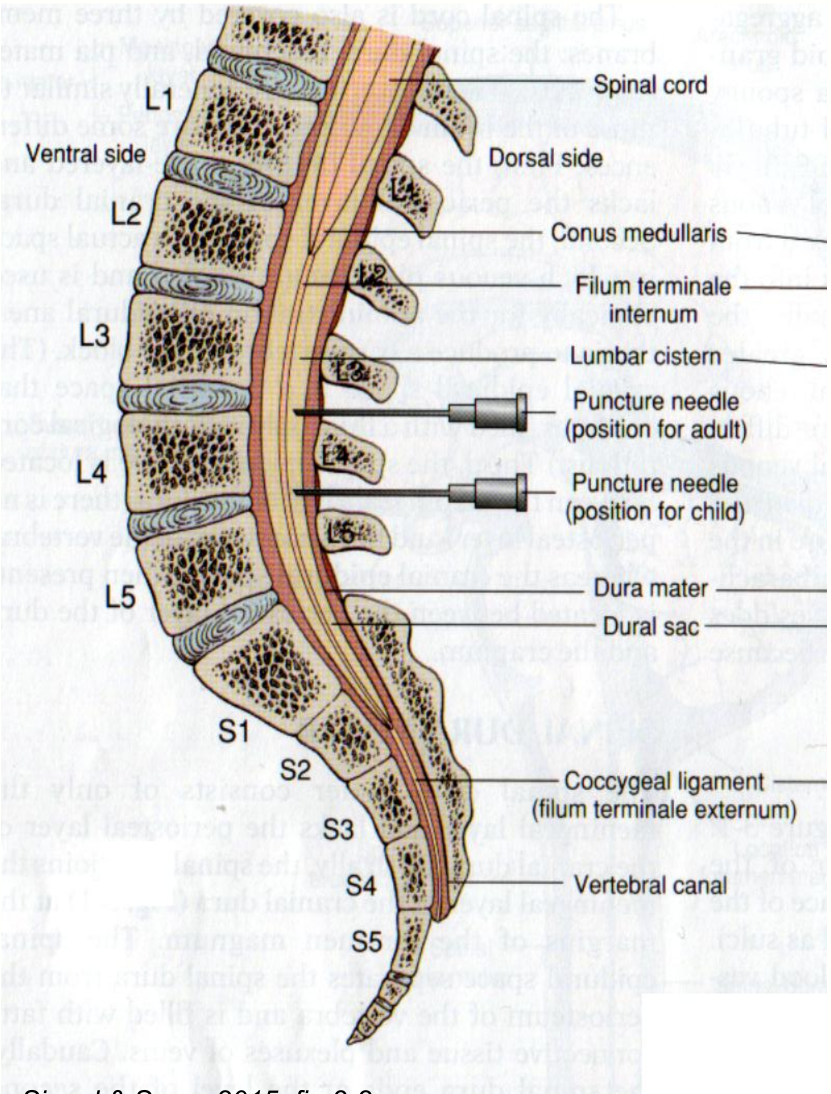
Lumbar Puncture *(Spinal Tap)*

- Location of spinal tap
  - Adults: L3/L4
  - Children: L4/L5
- Diagnostic purposes
  - Estimate of ICP
    - Normal ICP: 5 - 15 mm Hg



Siegel & Sapru 2015, fig 3-2

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Pathological Elevation of ICP

- ICP measurement
  - Ventricular catheter (most accurate)
  - Subdural screw/bolt
  - Epidural pressure sensor (no drainage option)
- Pathologically elevated ICP: > 15 mmHg (200 mmH_2O)
- Adverse effects of a rise in ICP
  - Nausea
  - Increased blood pressure (systemic hypertension)
  - Bradycardia
  - Papilledema

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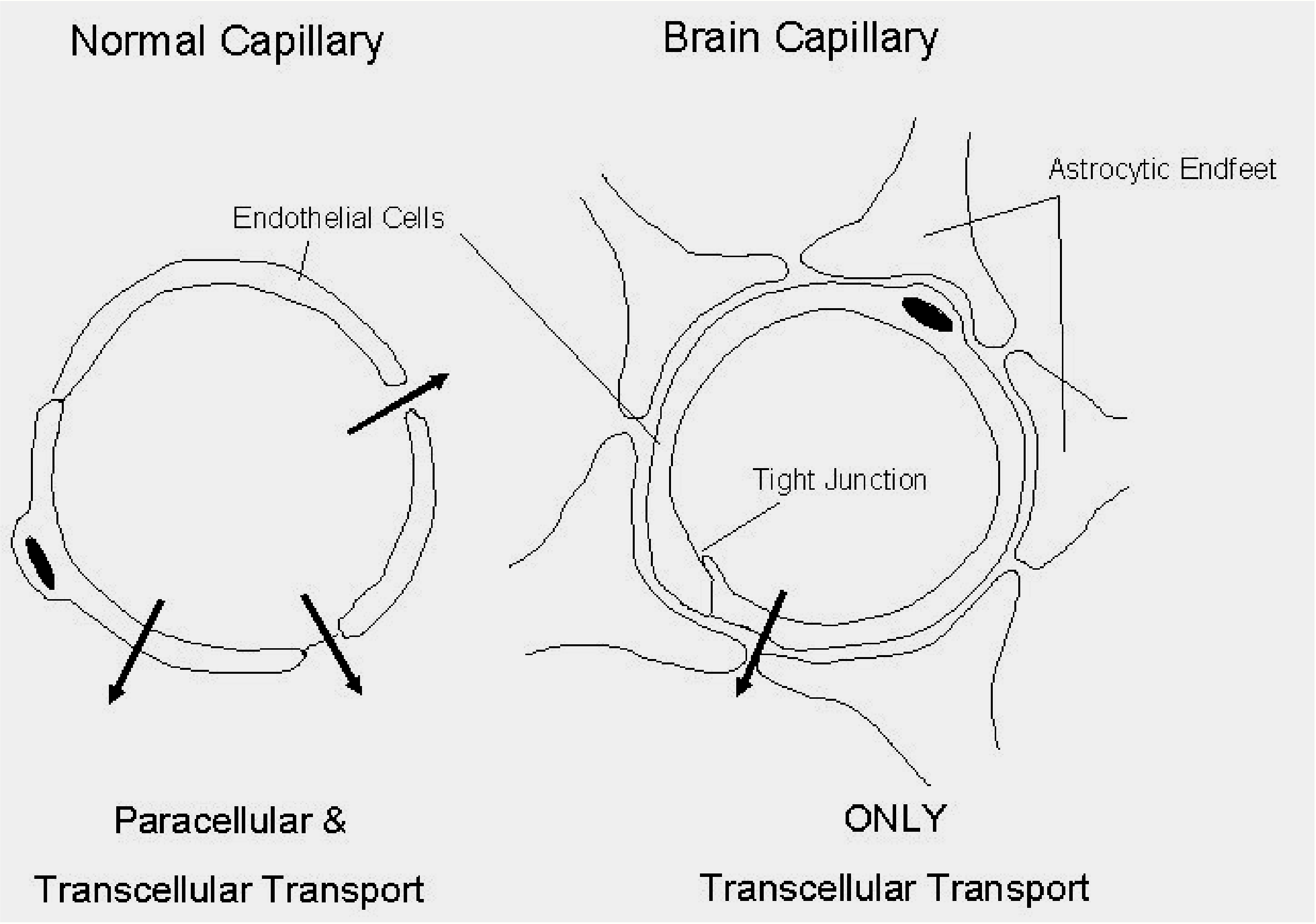
The Blood Brain Barrier

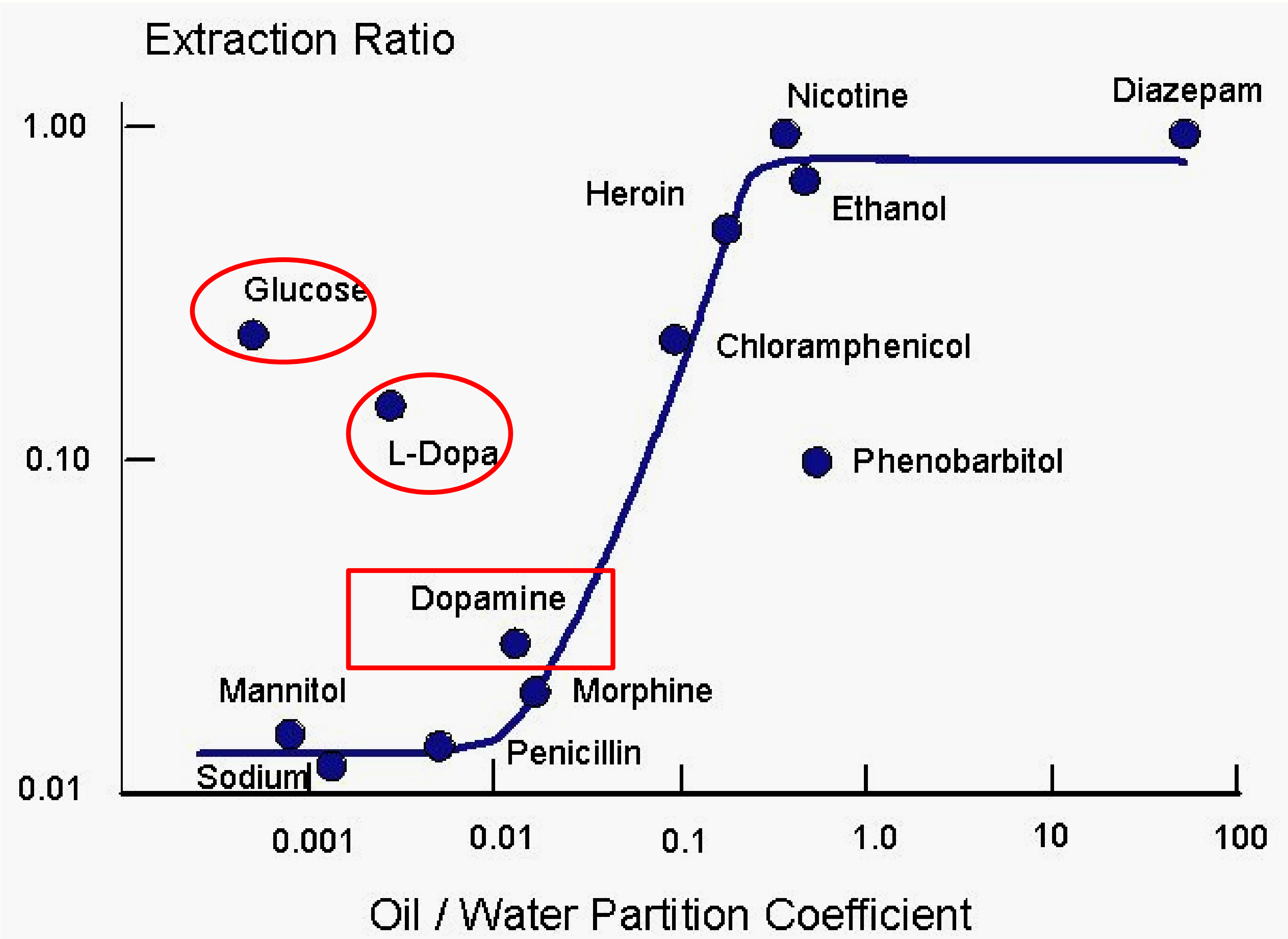
Tight Junctions between Endothelial cells

Astrocytic end feet



BBB-Main Transport: Transcellular Route

Diffusion rate across the bilayer depends on lipid solubility



NSCI.0531

https://neupsykey.com/the-blood-brain-barrier-choroid-plexus-and-cerebrospinal-fluid/

The Blood–Brain Barrier Is Permeable in Three Ways

Normal development and brain function require many substances that must cross brain microvessels. Entry into the brain is achieved primarily in three ways: (1) by diffusion of lipid soluble substances, (2) by facilitative and energy-dependent receptor-mediated transport of specific water-soluble substances, and (3) by ion channels.

Diffusion of Lipid-Soluble Substances

The brain is separated from the blood only by the large surface of endothelial cell membrane (approximately 180 cm2/g in gray matter). This membrane permits the efficient exchange of lipid-soluble gases such as oxygen (O2) and carbon dioxide (CO2), an exchange limited only by the surface area of the blood vessels and by cerebral blood flow. Barrier vessels are impermeable to molecules with poor lipid-solubility such as mannitol. The permeability coefficient of the blood–brain barrier for many substances is directly proportional to the lipid-solubility of the substance as measured by the oil–water partition coefficient (Figure D-6).

An example of the relationship of permeability and lipid-solubility is the correlation between the relative abuse potential of psychoactive drugs such as nicotine and heroin and their lipid-solubility. Increasing the lipid-solubility of drugs enhances their delivery to the brain. Drugs with great lipid-solubility, however, are poorly soluble in blood and bind to serum albumin protein, properties that reduce delivery to the brain. Lipid-solubility is not an accurate indicator of the permeability of some hydrophilic substances, such as glucose and vinca alkaloids, because of the presence of selective endothelial transport or enzyme systems that increase or inhibit permeability.

Because the paracellular route is closed, substances must pass via the **transcellular route.** Most of the cell membrane is occupied by the phospholipid bilayer component. Transport across this bilayer commonly depends on the solute's solubility in lipid.

Solutes that readily dissolve in lipid will cross the endothelial cell membranes. The oil/water partition coefficient expresses the degree to which solutes will move into lipid from an aqueous environment. The higher the coefficient is, the more effective the transfer from water to lipid will be. Small hydrophobic molecules, blood gases, small uncharged polar molecules, water, urea and glycerol all diffuse readily across phospholipid bilayer.

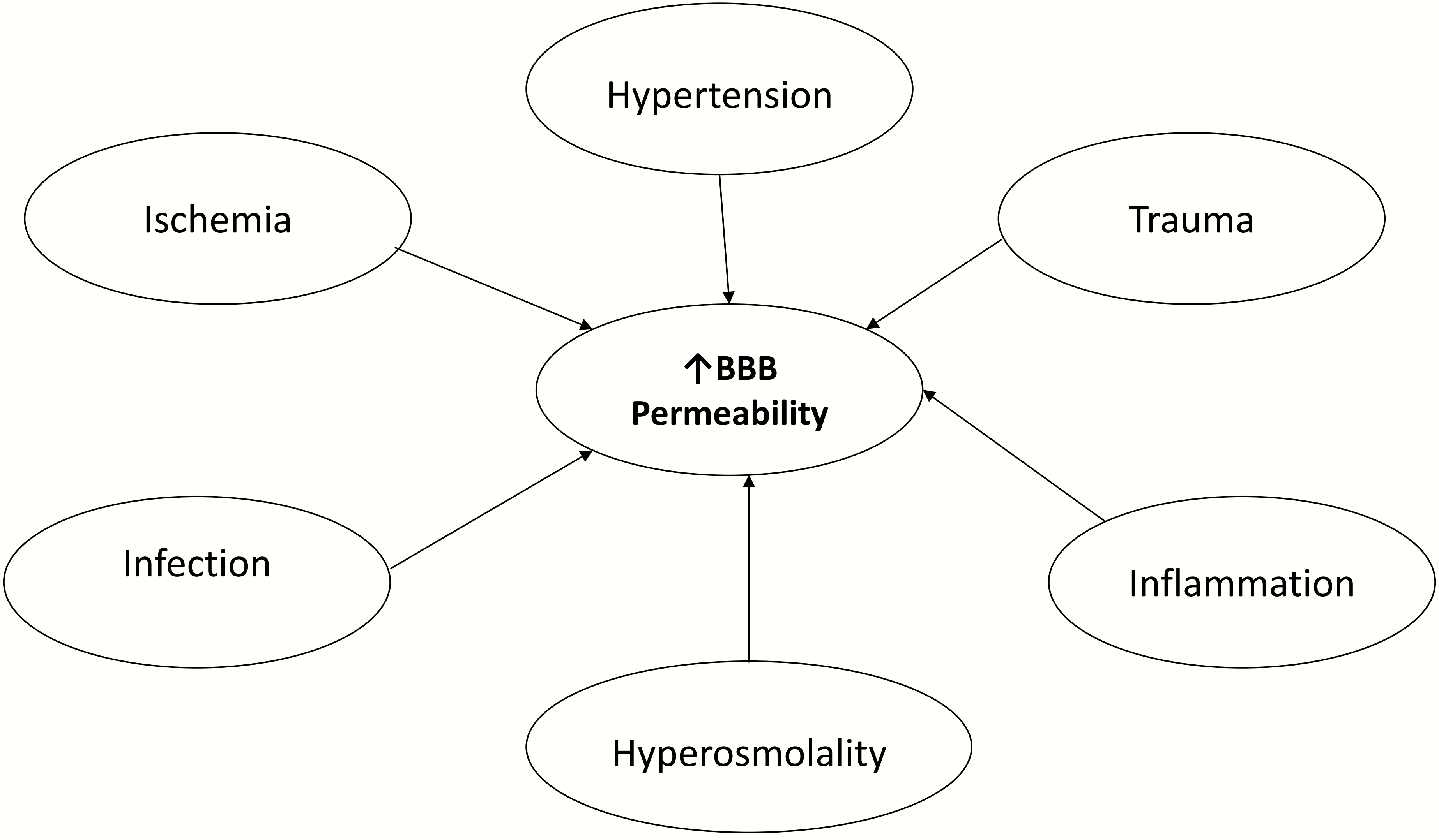
The line in the **graph above shows the expected relative extraction of solute from cerebral blood into brain tissue as a function of the partition coefficient.** Note that glucose and L-dopa do not lie on the curve. Glucose is transported into the brain via GLUT1 transporters (facilitated diffusion) and L-Dopa through a neutral amino acid carrier.

Phenobarbitol, while highly lipophilic, has a relatively low extraction ratio. This is because the substance is pumped out of the brain by an ATPase-dependent transporter.

Transport Mechanisms across the BBB

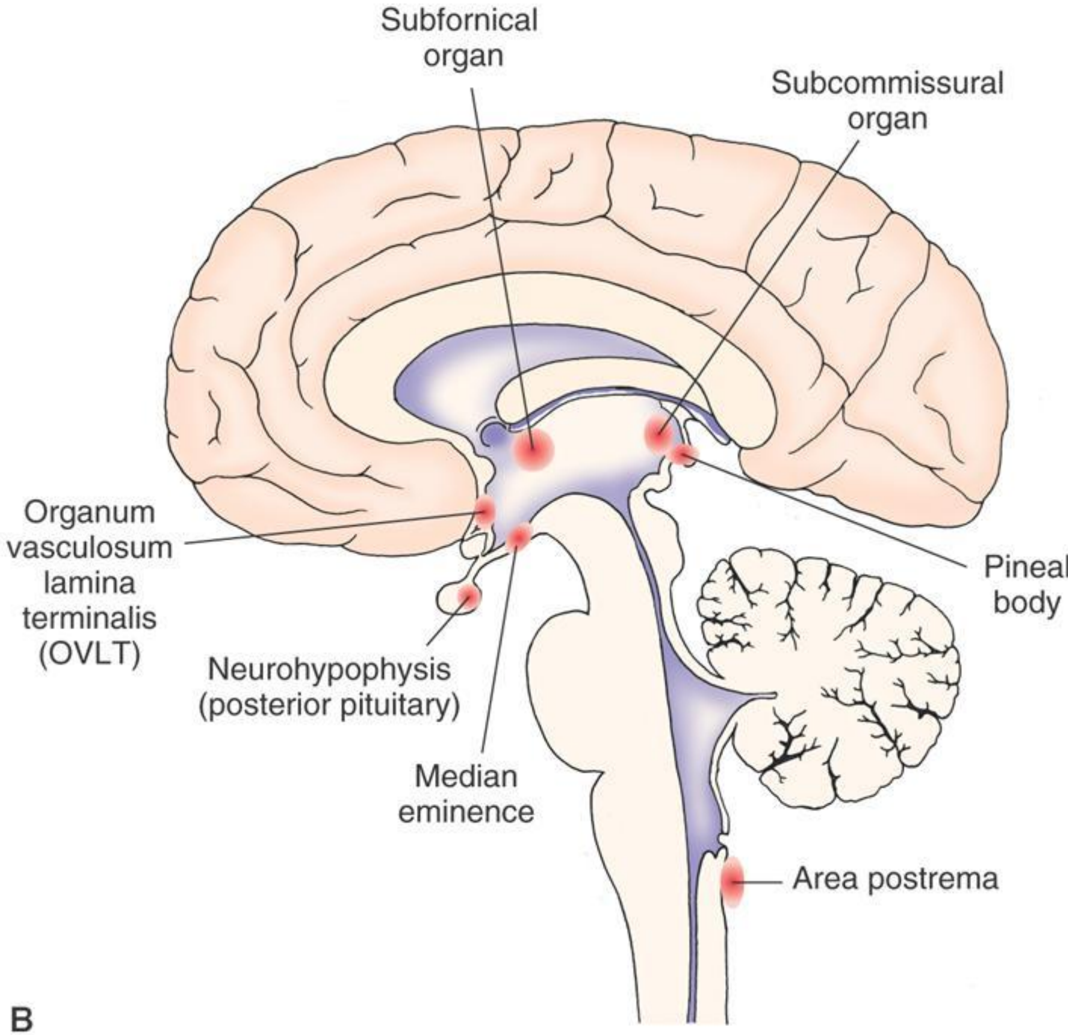
- Diffusion, facilitated diffusion and active transport
  - Lipid soluble molecules diffuse across the BBB via the phospholipid part of the endothelial cell membranes
  - Glucose crosses the BBB by facilitated diffusion via a transporter GLUT1 in the endothelial cell membranes
  - L-dopa (dopamine precursor) also crosses the BBB by facilitated diffusion, via a carrier for neutral amino acids
  - Glycine crosses the BBB (from brain to blood) via secondary active co-transport with Na^+, causing transfer of glycine in a Na^+-dependent fashion

Causes of Increasing BBB Permeability



Circumventricular Organs

- Brain regions where capillary endothelial cells lack tight junctions
- Regions where access of neurons to blood plasma is crucial for function (sensory or secretory)
    - E.g., blood pressure and osmolarity detection by the subfornical organ



Siegel & Sapru 2015, Fig 3.3b

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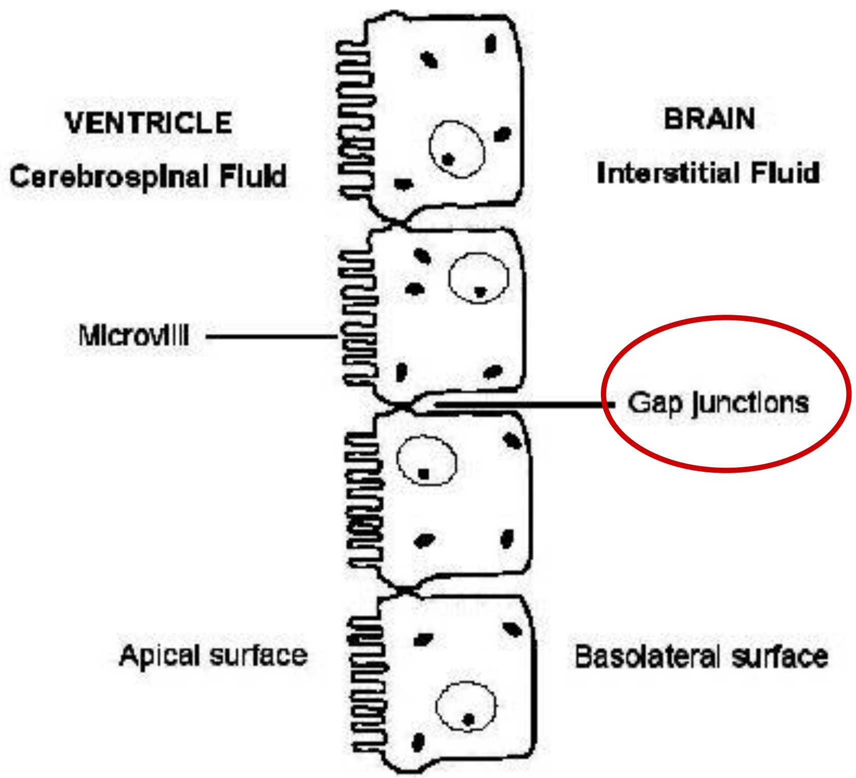
Secretion and Circulation of CSF

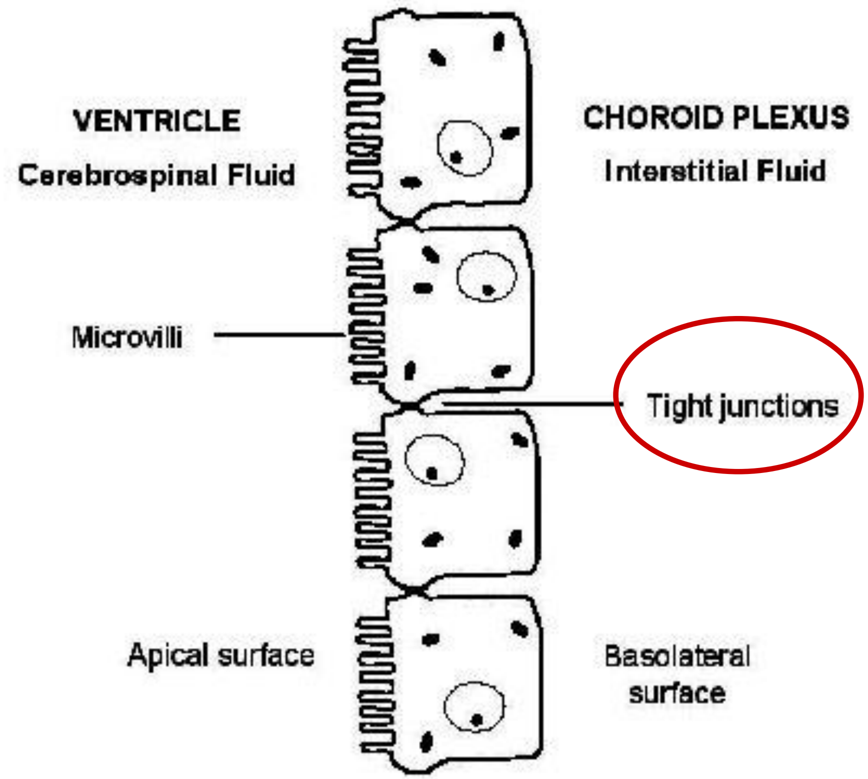
The Ventricular Lining is not Uniform in Structure and Function.

Extra choroidal Ependymal Cells lining Ventricles



Paracellular transport of CSF into interstitial space

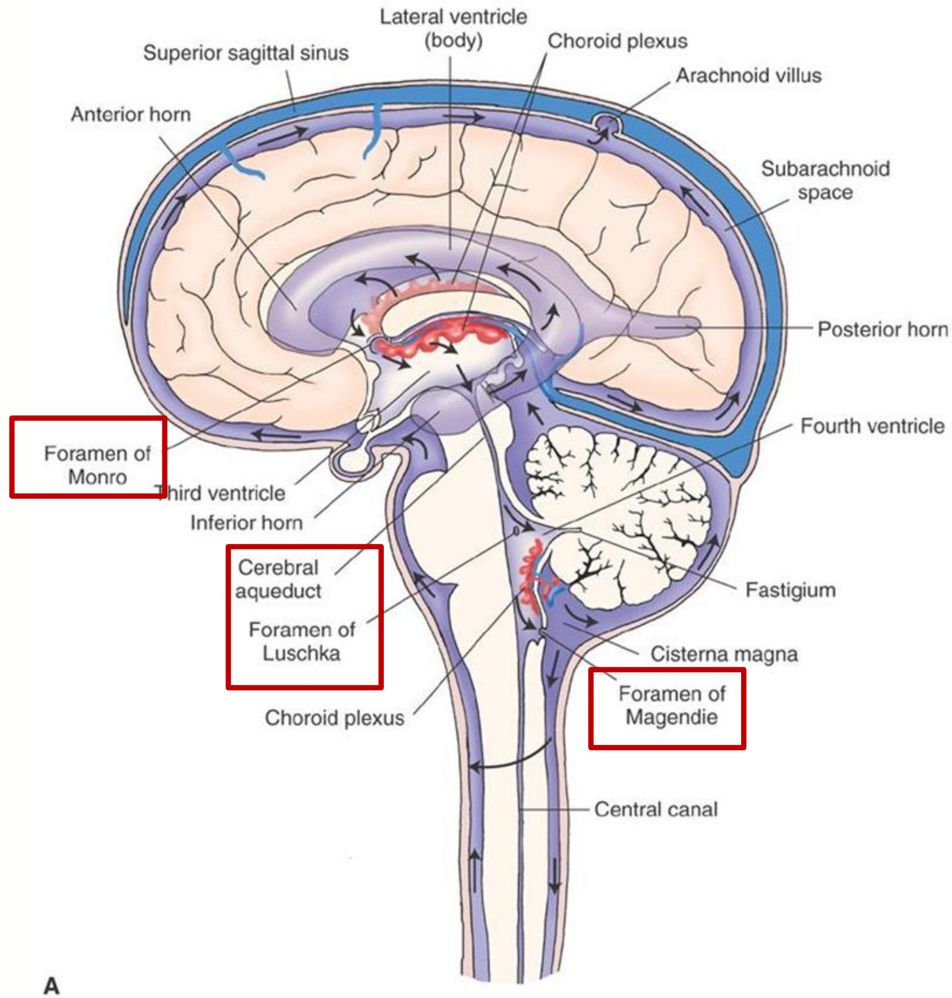
Secretory Epithelium of Choroid Plexus



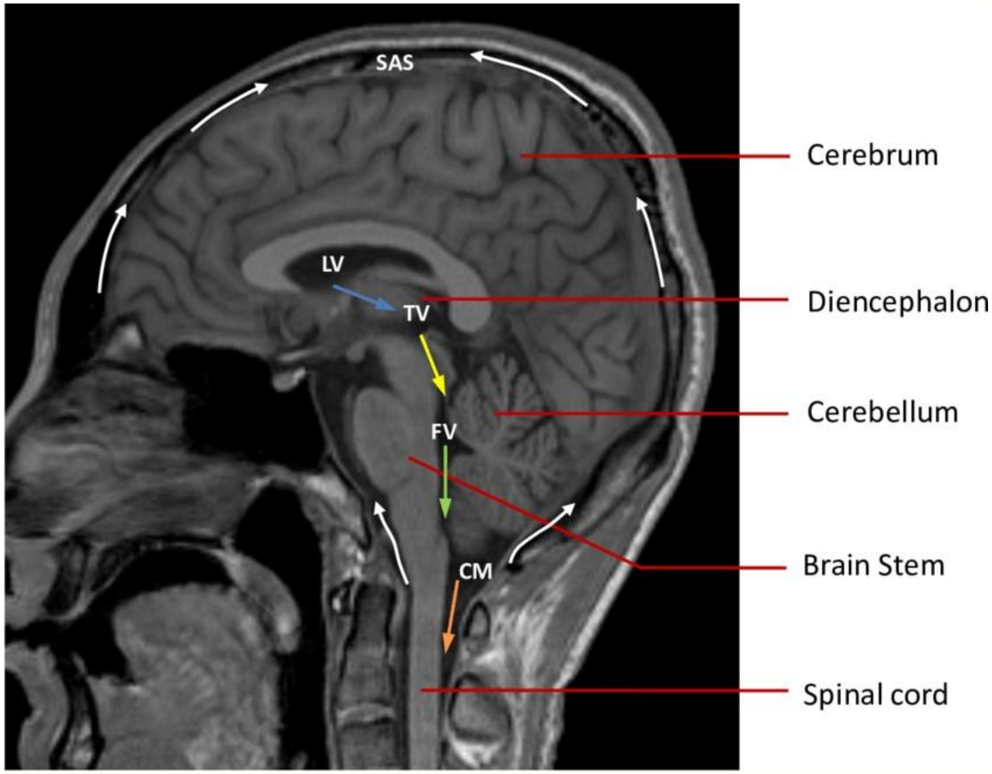
Transcellular CSF secretion,
pumps and transporters,
H_2O through aquaporin channels

CSF is Secreted, Circulated and Absorbed

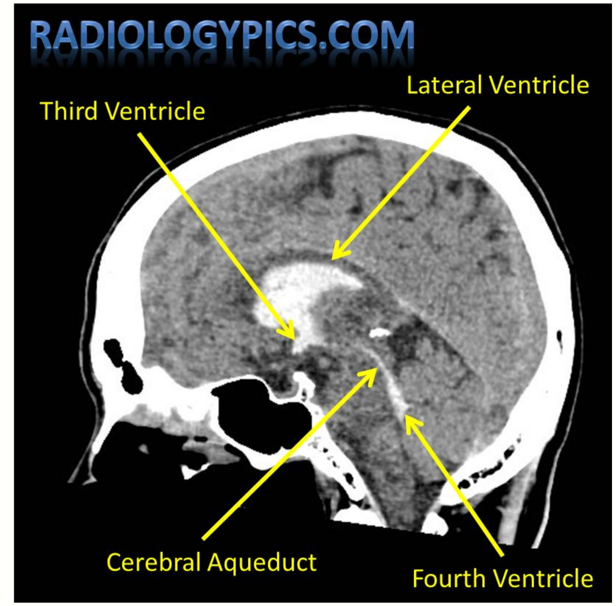
- CSF is absorbed by arachnoid villi
- CSF pressure > venous pressure
- Arachnoid villi act as one-way valves
- CSF production ~ 500 ml/day
- Total CSF volume ~ 140 ml
- Replaced ~ every 7 hrs



Imaging interpretation of brain ventricles



Sagittal T1 Weighted MR Image of a healthy adult brain showing the brain anatomy and typical flow of cerebrospinal fluid (CSF).



For identification of lateral and third ventricle please refer to the link.
https://www.radiologymasterclass.co.uk/gallery/ct_brain/ct_brain_stacks/ventricles_ct_brain

Li, Xiaogai. (2012). Finite Element and Neuroimaging Techniques to Improve Decision-Making in Clinical Neuroscience. 10.13140/RG.2.2.29290.93129.

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Diagnostic abnormalities of CSF

CHAPTER SUMMARY TABLE

Alterations in Cerebrospinal Fluid Composition in Some Pathological Conditions

Pathological Condition	Protein	Glucose	Cells
Subarachnoid hemorrhage	Increased (+)	Normal	Presence of RBCs
Guillain-Barré syndrome	$\boxed{\text{Increased (++)}}$	Normal	Presence of a few WBCs
Metastatic cancer in the menin-ges	Increased (+)	Normal or decreased	Presence of increased number of WBCs (lymphocytes) Tumor cells
Viral meningitis	Increased (+)	Normal	Presence of excessive number of WBCs (lymphocytes)
Tubercular meningitis	Increased (+)	Decreased	Presence of increased number of WBCs (lymphocytes)
Bacterial meningitis	Increased (+)	Decreased	Presence of increased number of WBCs (polymorphonuclear leuko-cytes)

CSF = cerebrospinal fluid; RBC = red blood cell; WBC = white blood cell. Plus sign (+) indicates increase. Relatively greater increase is shown by two plus signs (++).

NSCI.0066

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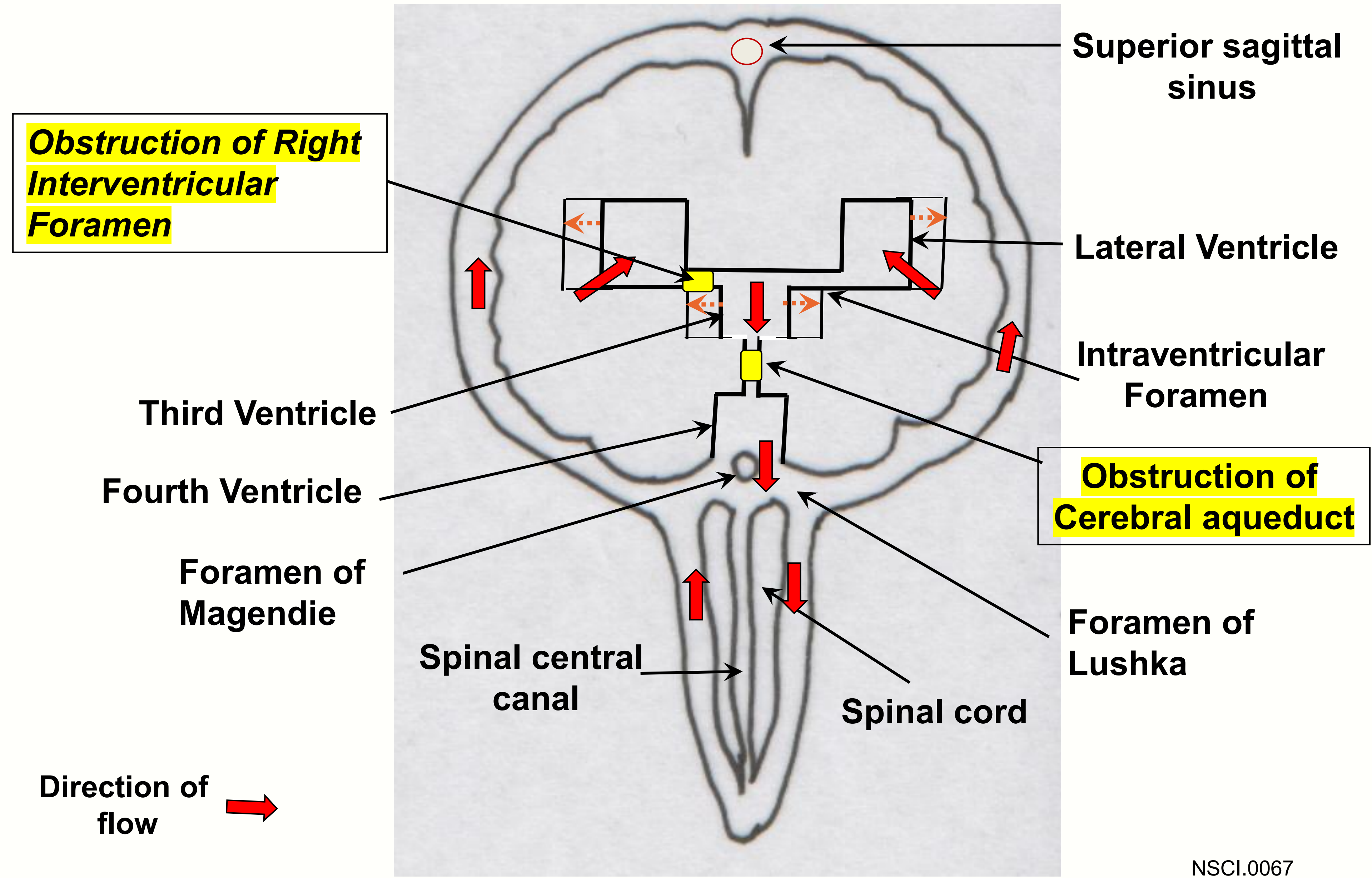
Non-communicating Hydrocephalus



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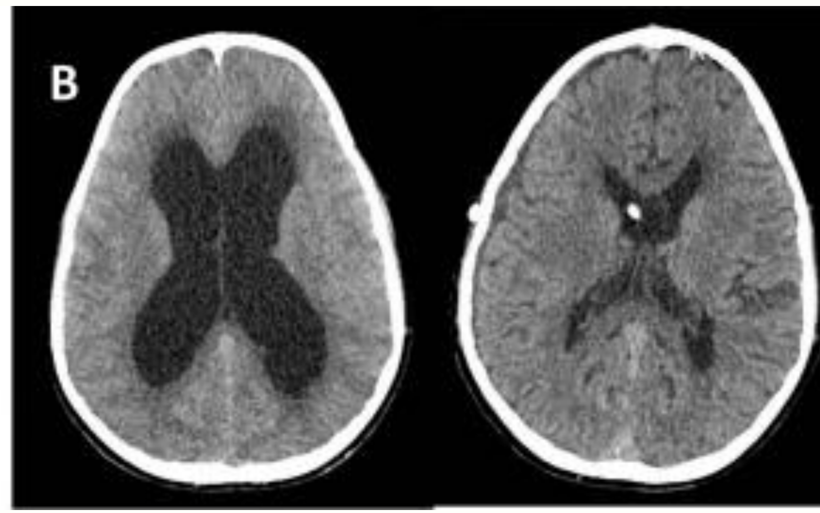
Non-communicating Hydrocephalus

Congenital Hydrocephalus: Obstruction of Cerebral Aqueduct



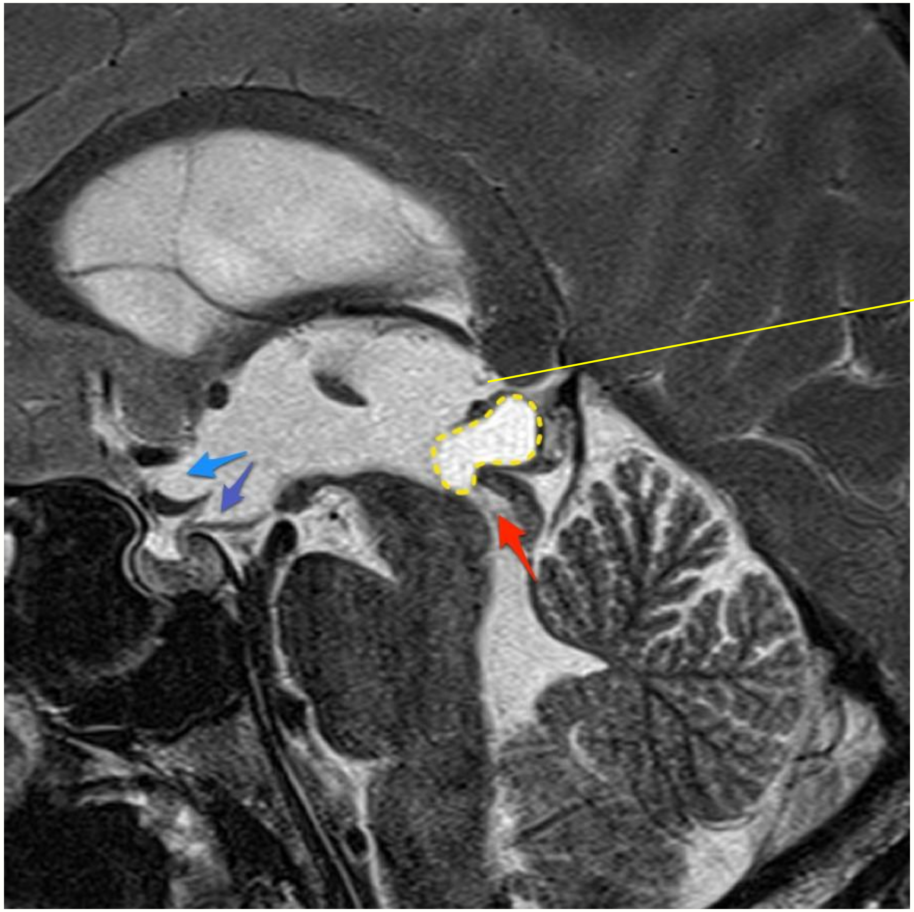
Before VP After VP

Ventriculoperitoneal (VP) shunting: brain ventricle and the other end in the abdominal cavity

The Open Neuroimaging Journal, 2018, 12, 10-15

Source: *Abrahams' and McMinn's Clinical Atlas of Human Anatomy*
Abrahams, Peter H., MB BS, FRCS (Ed), FRCR, DO (Hon) FHEA; Spratt, Jonathan D., MA (Cantab), FRCS (Eng), FRCR... Published January 1, 2020. Pages 1-85. © 2020.

Example: obstructing aqueduct



inferior bowing of the floor of the third ventricle

https://radiopaedia.org/cases/pineal-cyst-obstructing-aqueduct

A cystic pineal region mass (yellow dotted line) has prolapsed anteriorly and obstructed the superior aspect of the aqueduct (red). Hydrocephalus is present with inferior bowing of the floor of the third ventricle and ballooning of the supraoptic recess (blue) and infundibular recess (purple).

- **Noncommunicating (obstructive) hydrocephalus** is the most frequent type due to blockage of the flow of CSF within the ventricular system. In this case CSF is unable pass between ventricles or to exit the ventricular system into the subarachnoid space, due to obstruction of interventricular foramina, cerebral aqueduct or outflow foramina of fourth ventricle.

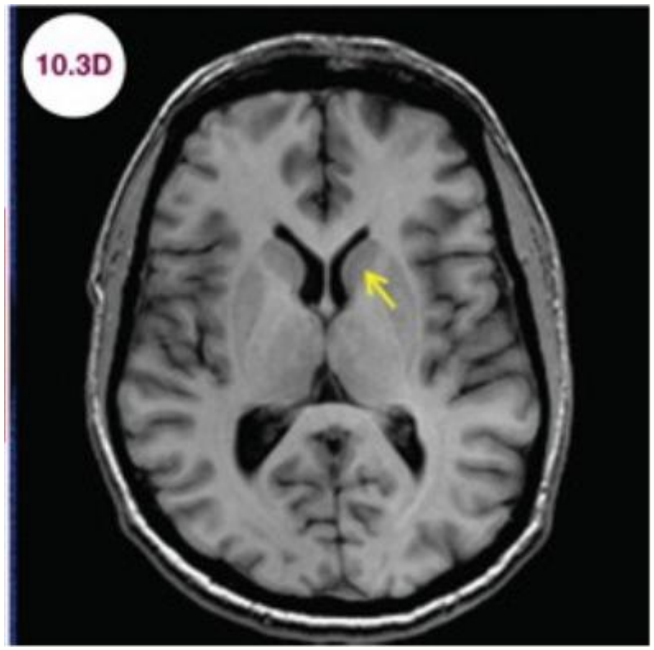
- In Infant: Increased CSF pressure →enlargement of the ventricles by expanding the skull because the cranial sutures are not yet closed. →large head size.

- The symptoms of hydrocephalus in infants include seizures, sleepiness, irritability, and downward deviation of the eyes (“sun setting” of the eyes).

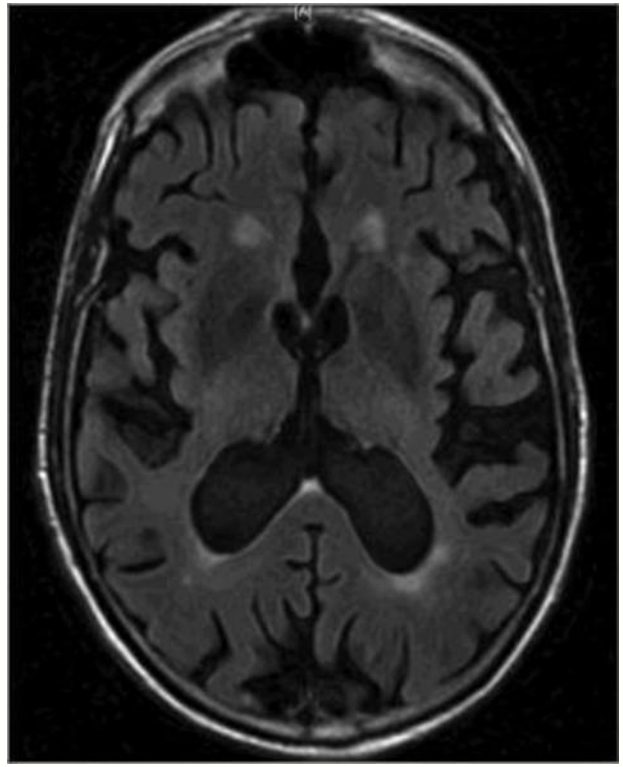
- in older children and adults include headache, nausea, vomiting, vision problems, sun setting of the eyes, difficulty in maintaining balance, problems with gait, poor coordination, urinary incontinence, lethargy, drowsiness, irritability, and changes in personality or cognition, including memory loss.

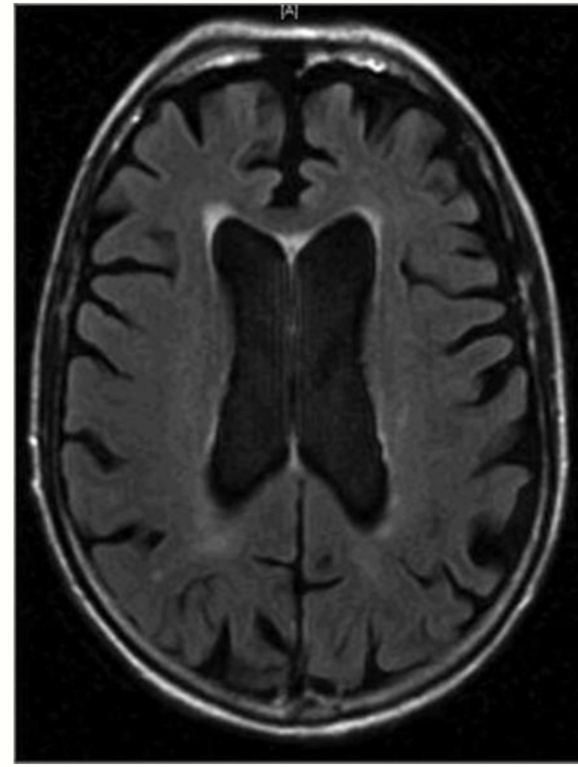
Hydrocephalus Ex Vacuo

Normal



https://neupsykey.com/neurodegenerative-diseases-2/





Case courtesy of A.Prof Frank Gaillard, Radiopaedia.org, rID: 9373

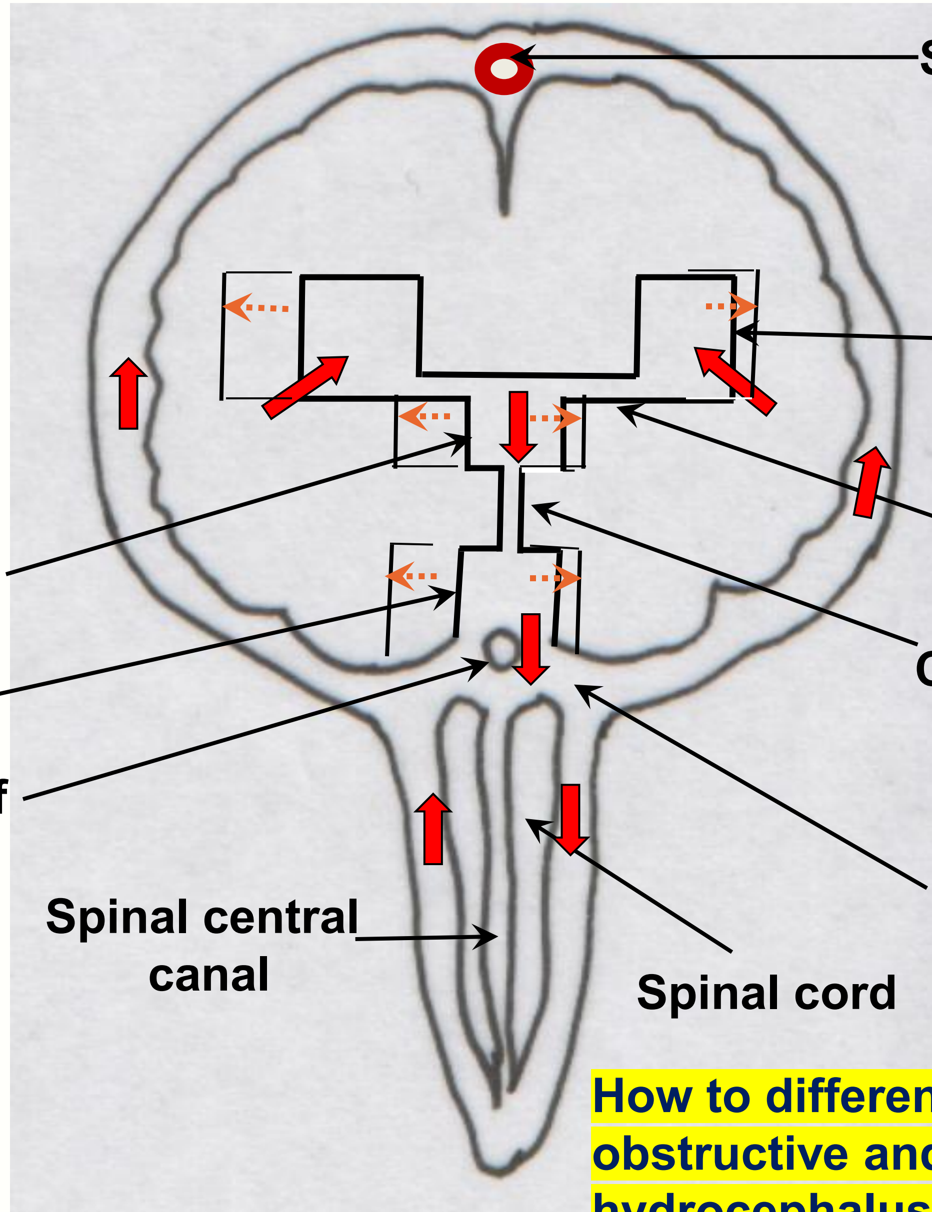
*** Brain tissue atrophy, increased CSF occupancy**

Hydrocephalus ex vacuo is associated with neurodegenerative disease. Cellular loss, as is seen in Alzheimer's and Huntington's diseases, permits enlargement of ventricles, as CSF passively occupies ever-increasing ventricular volume.

Communicating Hydrocephalus

Mismatched Absorption and production of CSF

Enlargement of all ventricles



Superior sagittal sinus

Third Ventricle

Intraventricular Foramen

Fourth Ventricle

Cerebral aqueduct

Foramen of Magendie

Foramen of Lushka

Spinal central canal

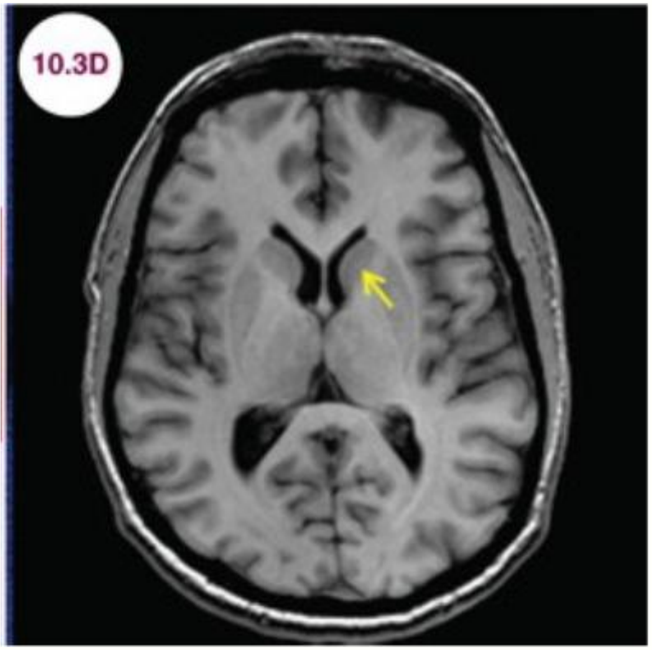
Spinal cord

Direction of flow

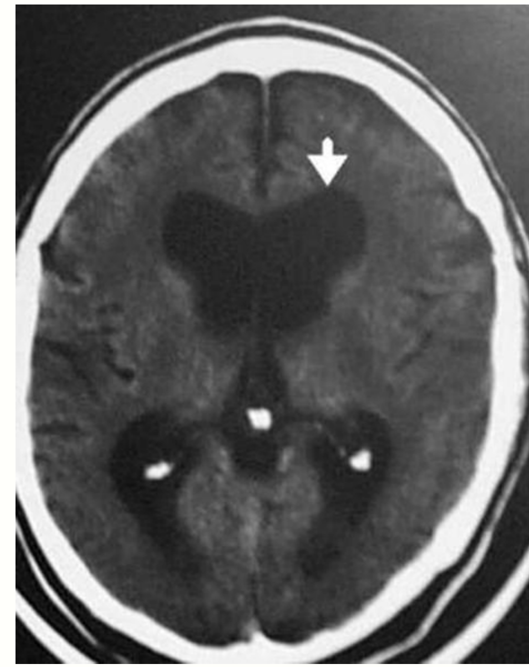
How to differentiate between obstructive and communicating hydrocephalus?

Communicating Hydrocephalus with Normal ICP

Normal



https://neupsykey.com/neurodegenerative-diseases-2/



Case courtesy of A.Prof Frank Gaillard, Radiopaedia.org, rID: 10241

- Episodic increase in ICP
- Expansion of ventricles distorts brain tissue
- Affects mainly the elderly

Communicating hydrocephalus can arise from impaired absorption of CSF by the superior sagittal sinus. This may reflect accumulations of cells in the subarachnoid space that impede passage of CSF through the arachnoid villi.

A rare form of **communicating hydrocephalus** results from tumors of choroid plexus (papillomas) with excess CSF secretion unmatched by rate of CSF drainage. Excess CSF secretion causes enlargement of all ventricles.

In noncommunicating hydrocephalus, a tracer dye injected into the lateral ventricle does not appear in the lumbar CSF, indicating that there is an obstruction to the flow of CSF in the ventricular pathways. If the movement of the CSF into the dural venous sinuses is impeded or blocked by an obstruction at the arachnoid villi, hydrocephalus developed in this manner is called a communicating hydrocephalus (nonobstructive hydrocephalus). In communicating hydrocephalus, a tracer dye injected into the lateral ventricle appears in the lumbar CSF, indicating that there is no obstruction to the flow of CSF in the ventricular or extraventricular pathways.

Lecture objectives

SOM.MK.I.BPM2.1.NB.1.NSCI.0061 List the fluid compartments in the CNS

SOM.MK.I.BPM2.1.NB.1.NSCI.0529 List the functions and constituents of the CSF

SOM.MK.I.BPM2.1.NB.1.NSCI.0530 Identify the typical sites for lumbar puncture in the adult

SOM.MK.I.BPM2.1.NB.1.NSCI.0062 Discuss principles of measurement and pathological elevation of intracranial pressure

SOM.MK.I.BPM2.1.NB.1.NSCI.0531 Describe the blood brain barrier (BBB) and its transport mechanisms

SOM.MK.I.BPM2.1.NB.1.NSCI.0063 Discuss weakenings of the BBB tied to the circumventricular organs

SOM.MK.I.BPM2.1.NB.1.NSCI.0064 Describe the role of the choroid plexus in CSF formation and of the arachnoid villi in CSF drainage

SOM.MK.I.BPM2.1.NB.1.NSCI.0065 Describe transfer of solute and water across barriers in the central nervous system

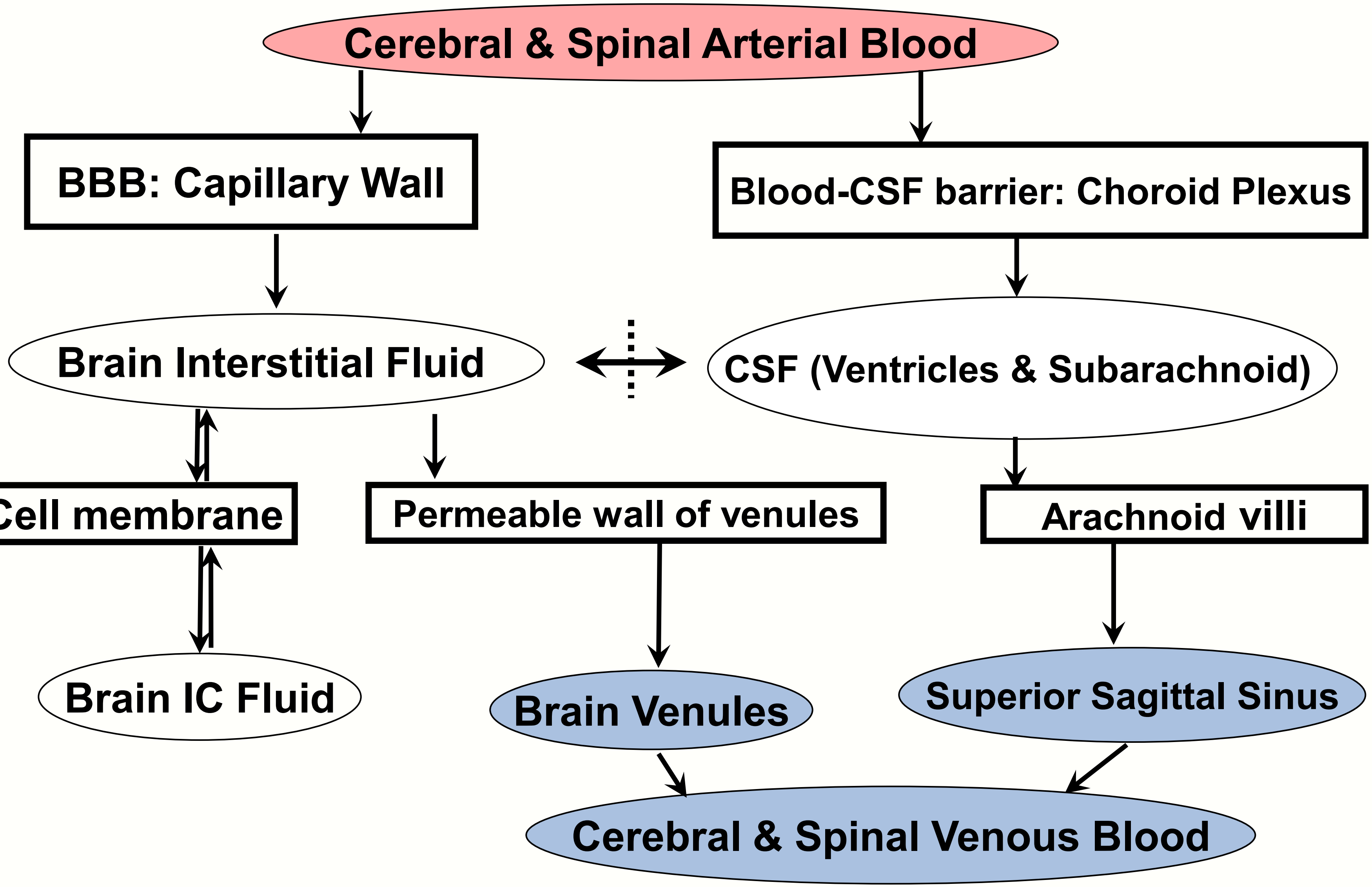
SOM.MK.I.BPM2.1.NB.1.NSCI.0066 Correlate composition of the CSF with disease processes

SOM.MK.I.BPM2.1.NB.1.NSCI.0067 Identify the causes of non-communicating and communicating hydrocephalus

SOM.MK.I.BPM2.1.NB.1.NSCI.0068 Describe pathways for movement of fluid and solutes among different compartments in the CNS

SOM.MK.I.BPM2.1.NB.1.NSCI.0069 Identify the causes of brain edema, other than hydrocephalus

Solute and Water Transport between Blood and Brain



NSCI.0068

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Brain Edema

Increased CNS fluid volume and/or disturbance of blood supply.
Osmotherapy is a key treatment.

Feature	**Vasogenic**	**Cytotoxic**
Origin	Increased capillary permeability	Cellular swelling
Location	White matter	Grey and white matter
Composition of edema fluid	Plasma filtrate (with plasma proteins)	Increased intracellular water and Na^+
Capillary permeability to large molecules	Increased (compromised BBB)	Normal (Intact BBB)
Clinical disorders	Brain tumor, abscesses, trauma, hemorrhage	Hypoxia, water intoxication, ischemia

Edema is an accumulation of fluid in the EC or IC space within the brain. **Disturbances of blood supply that interfere with the normal well-regulated exchange of solutes and water between blood and brain, can cause edema. Edema may be local or general.** Brain edema may be vasogenic or cytotoxic.

Vasogenic edema arises from damage to brain capillaries rendering them more permeable than normal, compromising the BBB, and allowing fluid to accumulate in the white matter and runs along axonal tracts (increase in IC fluid compartment). This could arise due to trauma or focal inflammation. Tumor-facilitated release of vasoactive and endothelial destructive compounds (e.g. arachidonic acid, excitatory neurotransmitters, eicosanoids, bradykinin, histamine, and free radicals).

Cytotoxic edema arises while the BBB remains intact and is caused by an inadequate blood supply to neurons and glia. When the Na/K pumps are deprived of ATP, the ionic gradients dissipate, and the cells swell with ensuing damage. Water intoxication is another source of cytotoxic edema.

During edema, the astroglial cells are capable of regulating their cellular volume after the initial swelling, and thus may suffer less damage than the other CNS cells. This renders them capable of regulating the neuronal microenvironment during the onset of damage. The most common treatment is osmotherapy.

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